Bitemporal Football Database

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1 Domain and why time matters

A fictional football league: six clubs, 60 players, six head coaches and two seasons. Contracts start, end and restart, and the paperwork is sometimes late or wrong. A payroll clerk, a regulator and an auditor can all ask *"who was FC Nord's coach on 2 September 2025, as the database knew it on date T?"* and need different answers: Rusu as known on 5 September, Pop as known on 12 September, and Rusu again today, after the handover was corrected from 1 to 3 September. A table with only valid time cannot give all three.

2 Modeling decisions

Periods. Every version table has `valid_from`, `valid_to`, `tx_from` and `tx_to`, all half-open (`from <= x AND x < to`); the seasons follow the same rule. Valid time is `DATE`, since contracts change by the day; transaction time is `DATETIME(6)` in UTC.

Sentinels, not NULL. Open ends are `9999-12-31` and `9999-12-31 23:59:59.999999`, so every predicate stays a plain, indexable range. `TIMESTAMP` is not used: it ends in 2038 and cannot hold the sentinel.

Anchor + version. `agent`, `player`, `club`, `coach` and `season` are non-temporal anchors. The bitemporal tables are `contract_version` (the relationship), `coach_assignment_version` (the role), `player_value_version` (the numeric attribute) and `club_license_version` (the parent for I2). `calendar_month` and `query_params` are helpers, and `story_event` is a hand-written index of the events, not the data.

tx_registry. Each logical change writes one `tx_registry` row, and every version row it touches gets that `tx_id` and `tx_time`. An E9 change to a licence, ten contracts and ten values is therefore one transaction.

Invariants. `v_integrity_violations` checks I1 (no overlapping current beliefs per key), I2 (contracts covered by the union of the club's licence periods), I3 (one coach per club at a time) and that transaction intervals are well formed (I4). The immutability part of I4 is enforced by the A1 triggers.

There are no deviations from the modeling rules.

Volume	Required	Actual
Players	50	60
Clubs	6	6
Coaches	5	6
Current contract beliefs*	60	78
Version rows	120	218

*Open `tx_to`, any valid time; 56 are valid on 2026-10-15.

3 Write protocol: E3, E4 and E8

Each write sets the clock `@sim_now`, adds a `tx_registry` row, closes the old belief (only `tx_to` changes) and inserts the new one. `@SENT_DATE` and `@SENT_TS` are the sentinels. The A1 procedures follow the same steps, and the A1 triggers block every other write.

E3, late entry. On 10 September the clerk records that Pop took over on 1 September.

```
SET @sim_now := '2025-09-10 14:00:00';
INSERT INTO tx_registry (tx_time, author, reason)
VALUES (@sim_now, 'clerk', 'E3 late entry: K2 takes FC Nord');
SET @tx := LAST_INSERT_ID();

UPDATE coach_assignment_version SET tx_to = @sim_now
WHERE coach_id = 1 AND club_id = 1
  AND tx_to = @SENT_TS AND valid_to = @SENT_DATE;

INSERT INTO coach_assignment_version
  (coach_id, club_id, role_title, valid_from, valid_to, tx_from, tx_to, tx_id)
SELECT coach_id, club_id, role_title, valid_from, '2025-09-01', @sim_now, @SENT_TS, @tx
FROM coach_assignment_version
WHERE coach_id = 1 AND club_id = 1
  AND tx_to = @sim_now AND valid_to = @SENT_DATE;

INSERT INTO coach_assignment_version
  (coach_id, club_id, role_title, valid_from, valid_to, tx_from, tx_to, tx_id)
VALUES (2, 1, 'Head coach', '2025-09-01', @SENT_DATE, @sim_now, @SENT_TS, @tx);
```

E4, correction. On 15 September the handover is corrected to 3 September: both beliefs from 10 September are closed and the corrected pair is inserted.

```
SET @sim_now := '2025-09-15 11:00:00';
INSERT INTO tx_registry (tx_time, author, reason)
```

```
VALUES (@sim_now, 'auditor', 'E4 correction: handover was 2025-09-03');
SET @tx := LAST_INSERT_ID();

UPDATE coach_assignment_version SET tx_to = @sim_now
WHERE club_id = 1 AND tx_to = @SENT_TS;

INSERT INTO coach_assignment_version
  (coach_id, club_id, role_title, valid_from, valid_to, tx_from, tx_to, tx_id)
VALUES (1, 1, 'Head coach', '2025-07-01', '2025-09-03', @sim_now, @SENT_TS, @tx),
       (2, 1, 'Head coach', '2025-09-03', @SENT_DATE, @sim_now, @SENT_TS, @tx);
```

E8, retraction. The P6/CS Port contract was never true, so it is closed in transaction time with no replacement.

```
SET @sim_now := '2026-02-12 09:00:00';
INSERT INTO tx_registry (tx_time, author, reason)
VALUES (@sim_now, 'auditor', 'E8 retraction: P6 at CS Port was never true');
SET @tx := LAST_INSERT_ID();

UPDATE contract_version SET tx_to = @sim_now
WHERE player_id = 6 AND club_id = 2
  AND tx_from = '2026-02-10 10:00:00' AND tx_to = @SENT_TS;
```

4 Challenges and failed attempts

Close + insert in a trigger. The first A1 design did the whole protocol in an `AFTER UPDATE` trigger. MariaDB refuses with `ERROR 1442`: a trigger cannot modify the table that fired it. The close + insert moved into procedures, and the triggers only guard.

Parameterised AS OF. `FOR SYSTEM_TIME AS OF (SELECT known_at FROM query_params)` fails with `ERROR 1064`, and a view using a variable fails with `ERROR 1351`. The native views use literal timestamps instead, one per Q6 date.

Dropping the index for EXPLAIN. `ix_coach_asof` also backs the foreign key on `club_id`, so `DROP INDEX` fails (error 1553). The plan without it uses `IGNORE INDEX` instead.

A drifting EXPLAIN estimate. Right after loading, the indexed plan estimated 7 rows instead of 4. Closing a belief updates `tx_to`, which is in the index, and InnoDB keeps the old entries until its purge runs. `tools/explain_q6.sql` now waits for purge first.

5 Insights: what the engines would enforce

Oracle 23ai. Temporal Validity (`PERIOD FOR`, `AS OF PERIOD FOR`) covers valid time, and Flashback Data Archive (`AS OF TIMESTAMP`, `VERSIONS BETWEEN`) keeps an append-only transaction history, which is I4. I1–I3 would still need constraints or triggers.

SQL Server 2022. System-versioned tables stamp transaction time and protect the history table (I4), but there is no valid-time period, so I1–I3 stay manual.

PostgreSQL 18/19. `WITHOUT OVERLAPS` keys would enforce I1 and I3. A `PERIOD` foreign key enforces I2 only against a single parent period; our licences have a six-month gap, so I2 would still need a trigger. `FOR PORTION OF` (19) shortens E4 and E5, but transaction time still needs triggers or an extension.

MariaDB (A2). `PERIOD FOR` and `WITH SYSTEM VERSIONING` reproduce Q1, Q5 and Q6 with zero differences from the core views. The costs: the open transaction end is `2106-02-07 06:28:15.999999` instead of 9999, and backdated history needs the `system_versioning_insert_history` loading mode.

6 Query outputs and EXPLAIN

All outputs come from `05_tests.sql`.

Q0. 0 rows on the final data. Each test returns one row and is rolled back: an overlapping contract for P1 (I1), P51 at FC Sud during the suspension (I2) and a second FC Nord coach (I3).

Q1. P1 on 2026-10-15: FC Nord, 120,000, permanent, valid from 2026-07-01.

Q2. FC Nord as known on 2026-10-15: Rusu from 2025-07-01 to 2025-09-03 (64 days), then Pop from 2025-09-03 (407 days). A rolled-back test splits Pop's row into two adjacent periods, and Q2 still returns a single Pop row.

Q3. 26 events: 14 proactive, 8 late, 4 corrections. *Derived* is computed from the version rows alone and agrees with the authored code on all 26. Delay is the recorded date minus the valid date, in days.

Code	Derived	Table	Key	Valid	Recorded	Delay	Old value	New value
E3	Late entry	contract	player:56	2025-07-10	2025-07-25 10:00	15	NULL	FC Sud, 150000.00
E3	Late entry	contract	player:57	2025-07-15	2025-07-28 11:00	13	NULL	FC Sud, 170000.00
E3	Late entry	contract	player:58	2025-08-01	2025-08-12 09:30	11	NULL	FC Sud, 175000.00
E3	Late entry	contract	player:59	2025-08-05	2025-08-20 14:00	15	NULL	FC Sud, 195000.00
E3	Late entry	contract	player:60	2025-08-10	2025-08-25 16:00	15	NULL	FC Sud, 200000.00
E4	Correction	licence	club:6	2025-07-01	2025-09-05 09:00	66	conditional	full
E3	Late entry	coach	club:1	2025-09-01	2025-09-10 14:00	9	K1 Mihai Rusu	K2 Andrei Pop
E4	Correction	coach	club:1	2025-09-01	2025-09-15 11:00	14	Pop from 2025-09-01	Pop from 2025-09-03
E3	Late entry	coach	club:2	2025-10-01	2025-10-10 12:00	9	NULL	K3 Vlad Ionescu

Code	Derived	Table	Key	Valid	Recorded	Delay	Old value	New value
E4	Correction	contract	player:7	2025-07-01	2025-11-10 09:00	132	114500.00	150000.00
E2	Proactive	coach	club:3	2026-01-15	2026-01-05 10:00	−10	NULL	K4 Dan Marinescu
E2	Proactive	coach	club:4	2026-01-15	2026-01-05 10:00	−10	NULL	K6 Radu Stan
E2	Proactive	coach	club:6	2026-01-15	2026-01-05 10:00	−10	NULL	K1 Mihai Rusu
E2	Proactive	contract	player:12	2026-01-15	2026-01-10 10:00	−5	CS Port	Rapid Vest (loan)
E2	Proactive	contract	player:1	2026-07-01	2026-01-18 12:00	−164	FC Nord, 93,500	FC Nord, 120,000
E4	Correction	value	player:8	2025-07-01	2026-01-20 15:00	203	1690000.00	3200000.00
E2	Proactive	value	player:10	2026-03-01	2026-02-01 09:00	−28	2,330,000	2,800,000
E3	Late entry	value	player:30	2026-02-01	2026-02-15 14:00	14	5,300,000	6,100,000
E2	Proactive	contract	player:7	2026-07-01	2026-06-15 11:00	−16	free agent	FC Nord, 160,000
E2	Proactive	licence	club:6	2026-07-01	2026-06-20 09:00	−11	suspended	full
E2	Proactive	contract	club:6	2026-07-01	2026-06-20 09:00	−11	10 ended contracts	10 contracts from 2026-07-01
E2	Proactive	value	club:6	2026-07-01	2026-06-20 09:00	−11	10 values	10 values +10%
E2	Proactive	contract	player:2	2026-07-01	2026-06-20 10:00	−11	FC Nord	Universitatea Est
E2	Proactive	contract	player:17	2026-08-01	2026-07-20 09:00	−12	free agent	CS Port, 195,000
E2	Proactive	contract	player:46	2026-09-01	2026-08-20 09:00	−12	free agent	CSM Centru, 140,000
E2	Proactive	contract	player:27	2027-01-01	2026-09-01 09:00	−122	free agent	Universitatea Est, 135,000

Q4. For 2025/26, [2025-07-01, 2026-07-01), 18 contracts overlap the season only in part and 44 cover it throughout.

Q5. On 2026-02-11 12:00 the database believed 63 contract rows, including the wrong P6/CS Port row (believed from 10 to 12 February).

Q6. FC Nord's coach on 2025-09-02, at three transaction dates.

Known at	Coach	Valid period	Transaction period
2025-09-05	Mihai Rusu	[2025-07-01, 9999)	[2025-07-01, 2025-09-10)
2025-09-12	Andrei Pop	[2025-09-01, 9999)	[2025-09-10, 2025-09-15)
2026-10-15	Mihai Rusu	[2025-07-01, 2025-09-03)	[2025-09-15, 9999)

Q7. FC Sud has 7 active contracts in July 2025, 10 from August to December, and 0 from January to June 2026 (the suspension).

Q8. 15 gaps, one per E6 restart, found with `LAG`.

Player	Club	Previous to	Next from	Gap (days)
P27 Catalin Cernat	Universitatea Est	2026-02-01	2027-01-01	334
P17 Octavian Cretu	CS Port	2026-01-01	2026-08-01	212
P51–P60 (10 players)	FC Sud	2026-01-01	2026-07-01	181 each
P7 Eduard Neagu	FC Nord	2026-02-01	2026-07-01	150
P12 Iulian Toma	CS Port	2026-01-15	2026-06-01	137
P46 Nicolae Stoica	CSM Centru	2026-07-01	2026-09-01	62

EXPLAIN Q6. With the index, Q6 is a range scan on `(club_id, tx_to, valid_from)`; without it, a full scan of all 19 rows.

Plan	Type	Key	Rows	Extra
with `ix_coach_asof`	`range`	`ix_coach_asof`	4	Using index condition; Using where
`IGNORE INDEX`	`ALL`	`NULL`	19	Using where